THIRD YEAR LABORATORY

LASER TWEEZERS

1 Aims

- To investigate the phenomenon of laser tweezers.
- To measure the viscosity of simple and complex fluids.
- To quantify the trapping forces applied.

2 Objectives

1. To quantify Brownian motion of microscopic particles in solution.
2. To use Brownian motion to calculate the viscosity of the medium.
3. To perform force measurements on fat globules in water.
4. To measure the dependence of the spring constant versus the laser power for trapped particles.
5. To investigate surface effects on trapped particles.

3 Introduction

Optical tweezers use the fact that light is able to exert a force on matter. This is not intuitive and we don't see this in everyday life because the forces are too small. We know that quantum theory predicts that light carries momentum,

$$\boldsymbol{p} = \hbar \boldsymbol{k} \tag{1}$$

and therefore

$$|\boldsymbol{p}| = \frac{h}{\lambda} = \frac{E}{c} \tag{2}$$

The prediction actually predates quantum mechanics and radiation pressure was suggested by Maxwell's e.m. theory (1873):

"In a medium in which the waves are propagated there is a pressure in the direction normal to the wave and numerically equal to the energy contained in unit a volume."

But even earlier Kepler (1619) hypothesised this whilst trying to determine the nature of comet tails. Early experiments to observe and measure this radiation pressure were hindered by thermal (radiometric) effects. In 1901 (before the invention of the laser) two independent results were published by Lebedev and Nichols & Hull who observed radiation pressure by looking at the effect of arc lamps on thin vanes. Lasers became available in the 70s, providing much higher intensities. Arthur Ashkin at Bell Labs pioneered optical manipulation in its current form and was awarded the Nobel prize for physics in 2018.

Ashkin's key insight was to use transparent particles in a transparent medium and he made use of Fresnel reflection from the particle interfaces, but with no absorption. In his first paper (1970, [1]) he used a weakly focused laser beam (long focal length, low NA lens) and observed particle guiding; particles were drawn into the beam and travelled in the beam direction. The fact the particles were drawn into the beam led him to determine there was a gradient force caused by the intensity gradient across the beam.

A further key result was that by using two counter-propagating beams, particles could be trapped through a combination of radiation pressure (also now known as the scattering force) and the gradient force. The next year Ashkin did some work using a single beam and trapped a particle by directing the beam upwards and balancing radiation pressure against gravity [2]. He also used another beam coming in from the side to move the particle up or down within the trap. When the laser was turned off he noted that the particle returned to the equilibrium position without oscillating, which indicated that the system was over-damped.

In 1986 Ashkin, Dziedic, Bjorkholm and Chu (Nobel Laureate) demonstrated a new way to trap particles using a single laser beam [3]. They used a high numerical aperture microscope objective lens to trap a particle not only in the transverse direction, but also in the axial direction. In this way particles could be trapped with the beam pointing downwards against the force of gravity.

They demonstrated that they could trap particles in the range 25 nm to 10 μm and discussed how this could be used:

"*They also open a new size regime to optical trapping encompassing macromolecules, colloids, small aerosols, and possibly biological particles*".

In 1987 Ashkin published the first paper to trap viruses and bacteria using. Later in 1987 he trapped cells using a near IR laser that did not damage the cells due to the reduced absorption.

It is recommended that you look up the above references and familiarise yourself with the background to optical tweezers. There is also a comprehensive text book by Jones et al, "Optical Tweezers: Principles and Applications" ([4]) that is available as a pdf via the library.

4 Theory

4.1 Photon momentum

As we have seen the momentum of a photon is given by Equation 1

and therefore

$$|\boldsymbol{p}| = \frac{h}{\lambda_\circ} = \frac{E}{c} = \frac{h\nu}{c}, \tag{3}$$

where we have introduced the vacuum wavelength, $\lambda_\circ$.

In a medium of refractive index n, the wavelength reduces to $\lambda_\circ/n$, so the momentum of the photon *increases* in the medium. This is not what you might expect, and is still a hotly debated topic. The controversy is known as the Abraham-Minkowski controversy named after Abraham who claimed the momentum is $h\nu/nc$ and Minkowski who claimed it is $hn\nu/c$. From Equation 3, Minkowski appears to be correct, and this is what we will use. The controversy revolves around the fact that we cannot really define the momentum of a photon in a medium, since it is questionable whether the photon model actually holds up.

If we take light travelling in a medium of refractive index n incident on a perfectly reflecting mirror then the magnitude of the change in momentum of each photon, $|\Delta\boldsymbol{p}|$ will be

$$|\Delta\boldsymbol{p}| = \frac{2nh\nu}{c}. \tag{4}$$

For a light beam of power P, the number of photons per second, N, is

$$N = \frac{P}{h\nu}, \tag{5}$$

so the magnitude of the force on the mirror, $|\boldsymbol{F}|$, will be

$$|\boldsymbol{F}| = \frac{2nh\nu}{c} \times \frac{P}{h\nu} = \frac{2nP}{c}. \tag{6}$$

This gives rise to what we know as *radiation pressure*. If the mirror is not perfectly reflecting, then to calculate the force on the mirror we simply multiply Equation 6 by the reflection coefficient.

For laser tweezers, where we are dealing with particles that are often spherical and transparent, we to need sum over all the photon trajectories (rays) through, and reflected off, the particle. This can be very complicated, and calculated forces rarely match up with measured forces. To overcome this a trap efficiency, Q, is defined so that

$$|\boldsymbol{F}| = \frac{QnP}{c}, \tag{7}$$

where n is the refractive index of the medium in which the particle is immersed.

4.2 Mie versus Rayleigh

The size of the particle relative to the wavelength determines whether we are in the Mie regime or the Rayleigh regime. If

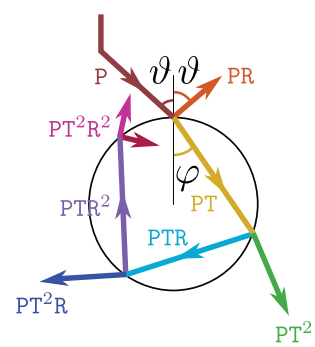

Figure 1: Ray model for a spherical particle. The incident beam of power P has reflection and transmission coefficients R and T at the interfaces respectively. ϑ is the angle the ray makes to the normal to the paticle surface. Taken from J. Sanders PhD thesis, 2012.

$$2r \leq \frac{\lambda}{\pi}, \tag{8}$$

where r is the particle radius, then we are in the Rayleigh regime and we have to treat the particle as a radiating dipole. In the opposite regime

$$2r >> \frac{\lambda}{\pi}, \tag{9}$$

then we are in the Mie regime and we can use the ray optics approach as above.

4.2.1 Mie regime

If we take a transparent spherical particle, then we can model the rays interact with the particle as in Figure 1. At each interface we need to consider both a reflected and a transmitted wave. The reflection and transmission coefficients will depend on both the angle and the polarisation, and can be calculated using the Fresnel equations.

Figure 2 shows the resultant forces due to the momentum transfer and we can see that the particle will always want to move towards the focus of the laser beam. In terms of the actual forces involved, we can split them into two. One is the scattering force, $\boldsymbol{F}_{scat}$, which is due to back reflection from the particle, and the other is the *gradient* force due to the gradient of the

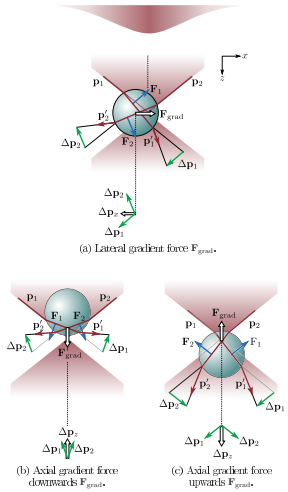

Figure 2: The forces on a particle in the Mie regime. (a), (b) and (c) are when the particle is displaced transversely, above and below relative to the focus respectively. Taken from J. Sanders PhD thesis, 2012.

electric field, $\boldsymbol{F}_{grad}$ (this acts in both axial and transverse direction). The total force is the sum of these two.

4.2.2 Rayleigh regime

In the Rayleigh regime the particle is much smaller than the wavelength and the ray method cannot be used to calculate the forces. We need to consider the induced dipole and the subsequent re-radiated light. The magnitude of the scattering force, $|\boldsymbol{F}_{scat}|$, is given by

$$|\boldsymbol{F}_{scat}| = \frac{I_\circ \sigma n_m}{c}, \tag{10}$$

where σ is the scattering cross section, $I_\circ$ the incident light intensity and n_m the refractive index of the medium. The scattering cross-section is given by

$$\sigma = \frac{128\pi^5 r^6}{3\lambda^4}\left(\frac{m^2-1}{m^2+1}\right)^2, \tag{11}$$

where r is the radius of the particle and m is the ratio of the refractive indices of the medium and particle,

$$m = \frac{n_p}{n_m}. \tag{12}$$

The strong dependence on wavelength due to the fourth power of λ in the denominator of Equation 11 explains why the sky is blue.

The magnitude of the gradient force, $|\boldsymbol{F}_{grad}|$, depends on the gradient of the intensity,

$$|\boldsymbol{F}_{grad}| = \frac{2\pi\alpha}{cn_m^2}\nabla I_\circ, \tag{13}$$

where the polarizability, α, of the particle is given by

$$\alpha = n_m^2 r^3 \left(\frac{m^2-1}{m^2+1}\right)^2. \tag{14}$$

Unfortunately most particles that are used for trapping fall between the Mie and Rayleigh regimes and therefore a more complete electromagnetic theory is required. However, Equation 7 can be used, because Q can accommodate any uncertainties in the theoretical model.

4.3 The trapping force

It is often very important to measure the trapping force, either to quantify the quality of the trap (e.g. for the optical alignment), and when the trapped particle is used to perturb its environment (e.g. the force applied on a molecule).

To model the trap we treat it as a Hookean spring, with the particle as an over-damped oscillator. We assume that for a small distance away form the centre of the trap the trapping force is proportional to the displacement i.e. it acts like a spring.

Considering one dimension, $x(t)$, the Langevin equation of motion

$$m\ddot{\boldsymbol{x}}(t) + \gamma\dot{\boldsymbol{x}}(t) + \kappa\boldsymbol{x}(t) = \boldsymbol{F}_B(t), \tag{15}$$

where m is the mass of the particle, γ is the viscous drag coefficient of the medium, κ is the optical trap stiffness and $\boldsymbol{F}_B(t)$ is the force due to thermal fluctuations. The first term in Equation 15 is the inertial force, and can be neglected since most trapping is in the low Reynolds number regime, $R_e << 1$, where viscous forces dominate.

4.3.1 Stoke's drag method

The viscous drag is given by Stoke's law

$$\gamma = 6\pi\eta r, \tag{16}$$

where η is the dynamic fluid viscosity. Hence the Stoke's viscous force, $\boldsymbol{F}_S$, (the second term in Equation 15) is

$$\boldsymbol{F}_S = \gamma\dot{\boldsymbol{x}}(t) = 6\pi\eta r\dot{\boldsymbol{x}}(t). \tag{17}$$

If the force due to thermal fluctuations is ignored (these are also very small, but we can use them later on), then Equation 15 becomes

$$\boldsymbol{F}_T = -\kappa\boldsymbol{x} = \boldsymbol{F}_S = \gamma\dot{\boldsymbol{x}}, \tag{18}$$

where $\boldsymbol{F}_T$ is the trapping force. Because the Stoke's force increases with velocity, one way to measure the trapping force is to trap a particle, then move the container in which it is trapped at ever increasing velocities until the Stoke's force exceeds the trapping force. At this point the particle will leave the trap and the velocity at which it does so will allow the Stoke's force (and hence the trapping force) to be calculated.

4.3.2 The equipartition method

Each degree of freedom *in thermal equilibrium* has $k_BT/2$ thermal energy. We can equate the thermal energy to that of a particle in a 1D optical trap (modelled as a harmonic potential), therefore

$$\frac{1}{2}k_BT = \frac{1}{2}\kappa\langle x^2\rangle \tag{19}$$

where $\langle x^2\rangle$ is the particle's positional variance. The probability, $P(x)$, of finding a particle at x within a potential $U(x)$ is described by a Boltzmann distribution as

$$P(x) \propto \exp\left(-\frac{U(x)}{k_BT}\right). \tag{20}$$

$P(x)$ is proportional to $H(x)$ which is a histogram of particle positions. We take the harmonic form

$$U(x) = \frac{1}{2}\kappa(x-x_\circ)^2 \tag{21}$$

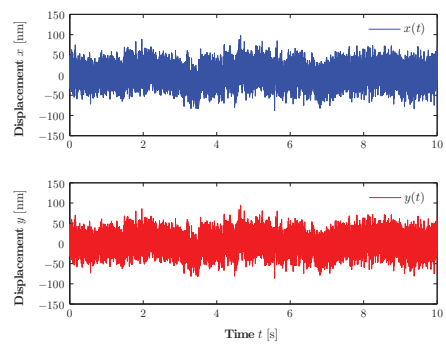

Figure 3: An example of positional vs time data for a trapped sphere.

and therefore

$$H(x) \propto P(x) = C\exp\left(-\frac{\kappa}{2k_BT}(x-x_\circ)^2\right). \tag{22}$$

This is the form of a Gaussian

Experimentally we measure the position as a function of time and calculate $H(x)$. We then fit a Gaussian to determine $\langle x^2\rangle$ and therefore κ. This *does not* require a knowledge of γ. See Figure 3 for an example of position data versus time for x and y. Figure 4 shows the corresponding histogram for this data with a Gaussian fit. Also shown is the calculated form of the potential for this data.

4.3.3 Autocorrelation Function (ACF) and Mean Square Displacement (MSD)

We can solve the over-damped Langevin Equation (15) for $x(t)$ to give the autocorrelation function (ACF). This decays exponentially and is given by

$$\langle x(t+\tau)x(t)\rangle = \langle x^2\rangle\exp\left(\frac{-\tau}{\tau_c}\right), \tag{23}$$

where τ is a delay known as the autocorrelation time and τ_c is the decay time

$$\tau_c = \frac{\gamma}{\kappa}. \tag{24}$$

The Mean Square Displacement (MSD) of $x(t)$ is given by

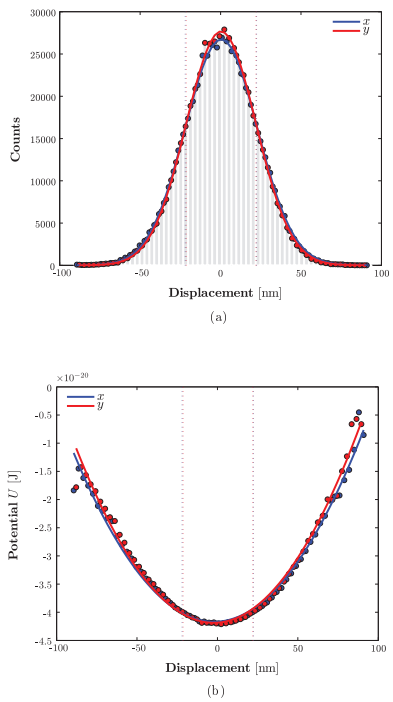

Figure 4: The same data in Figure 3 plotted as a histogram with a Gaussian fit (a), and the corresponding potential plot (b) as a function of displacement

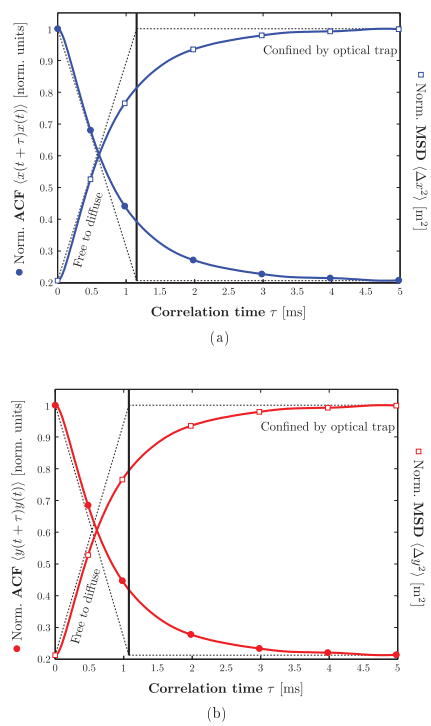

Figure 5: Autocorrelation function (ACF) and Mean Square Displacement (MSD) plots

$$\langle \Delta x^2(\tau) \rangle = \langle (x(t+\tau) - x(t))^2 \rangle. \tag{25}$$

This is related to the ACF by

$$\langle \Delta x^2(\tau) \rangle = 2\langle x^2 \rangle - 2\langle x(t+\tau)x(t) \rangle \tag{26}$$

and therefore

$$\langle \Delta x^2(\tau) \rangle = \frac{2k_BT}{\kappa}\left[1 - \exp\left(-\frac{\kappa\tau}{\gamma}\right)\right], \tag{27}$$

and so we can see that the ACF and MSD are complementary. See Figure 5 for example ACF and MSD plots.

Looking at Equation (23) and from Figure 5, when $\tau \to 0$, we just get $\langle x^2 \rangle$, i.e. free diffusion, and in this region the plot is approximately linear. For $\tau \to \infty$, the ACF$\to 0$ and the MSD $\to 1$, and the particle is confined in the trap.

To generate the MSD data, the position (along one axis, for example the x-axis) of a microsphere can be tracked using the camera. Since the camera records a sequence of frames equally spaced in time, the time averaged MSD after N frames can be determined from

$$\langle \Delta x^2(t_n) \rangle = \frac{1}{N-n}\sum_{i=1}^{N-n}(x_{i+n} - x_i)^2, \tag{28}$$

where $n = 1, ...N-1$ and the expression averages over all the displacements of duration $n\Delta t$. Due to the number of samples in the averaging process the MSD should be reasonably smooth at short times and have much more significant noise at longer times.

4.3.4 Free diffusion

To calculate the viscosity from the MSD data we can start with the Stokes-Einstein equation relating the diffusion coefficient, D, to the viscosity

$$D = \frac{k_BT}{\gamma} = \frac{k_BT}{6\pi\eta r}. \tag{29}$$

The diffusion coefficient is related to the MSD by

$$\mathrm{MSD}(\tau) = 2nD\tau, \tag{30}$$

where n is the number of dimensions, which in our case is 2.

Using Equations 30 and 29 we can see that

$$\frac{\mathrm{MSD}(\tau)}{\tau} = 2nD = \frac{4k_BT}{6\pi\eta r} = m, \tag{31}$$

where m is the gradient of a plot of MSD versus τ.

Hence the viscosity is given by

$$\eta = \frac{2k_BT}{3\pi r m}. \tag{32}$$

5 Carrying out the experiment

5.1 Equipment

The basis of the equipment is a laser tweezer kit (EDU-OT1/M) [5] from Thorlabs. Their website has a detailed list of the components in this kit, and there is a manual that includes details of possible experiments.

The basic laser tweezer kit comprises a collimated laser diode, which is passed through a high numerical aperture microscope objective lens to form a highly converging focussed optical trap. Before the objective lens the beam passes through a collimating lens, two beam expanding lenses, reflects off two mirrors and then goes through a beam splitter. A white light LED is used to illuminate the sample, with an image formed on a CMOS camera after the light passes through the beamsplitter. A small volume of liquid is contained in a well on a microscope slide. This sample can then be manipulated in all three axes using translation stages, two of which are moved using motors controlled from the computer, and the other via a micrometer (the axial direction). A schematic of the optical set is shown in Figure 6.

There are a number of pieces of software you will need. The camera is run through ThorCam, and the laser diode and the motorised translation stages through Kinesis. The Thorlabs manual explains the basics functions. You MUST NOT exceed the maximum current through the laser diode, and the translation stages cannot move negative. Consider setting the stages to around 1 mm before starting. Don't use the HOME button. You should spend some time familiarising yourself with how to control the laser and translation stages. Good control of the sample is essential for this experiment.

You will also need to use Trackmate in ImageJ (also known as Fiji) or TrackPy in Python to track the particles and hence generate x, y data for the particles' positions.

5.2 Samples

There are a number of different samples you can use for this experiment. The majority consist of small microspheres in solution. There are silica beads and polystyrene beads. You can also make up your own spheres of fat or oil in water using either cream or immersion oil. The density of the sphere (compared to the surrounding liquid), the size of the sphere, and the refractive index of the sphere can all affect measurements.

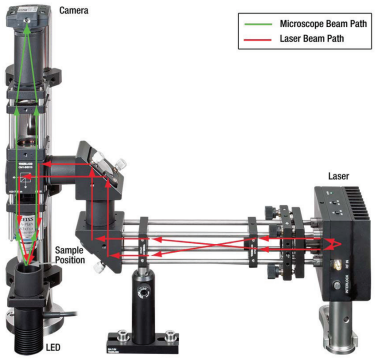

Figure 6: The optical paths for the laser and image formation (the sample stage is omitted for clarity).

To prepare a sample using the microscope slides with a well, with a cover slip is needed that has been sealed in place with nail varnish. Be careful not to contaminate the concentrated source of microspheres. You will need a very small quantity of microspheres, diluted down greatly. Label your slides carefully so that you can identify them.

5.3 Experiments

There are a number of experiments you can carry out. Suggestions include:

1. Investigate Brownian motion of microspheres of different sizes by measuring the MSD.
2. Use Equation 32 to evaluate the viscosity of the medium.
3. Tweezer particles and carry out force and/or spring constant measurements using the equipartition theory. This can be done as a function of laser power.
4. Look at the effects of trapping a particle close to a surface.

5.4 Extensions

Successful extensions have included:

1. Trap particles in standard foods (gels, mayonnaise, starches etc.) and calculate their viscoelesticity.

2. Create an oil/water interface and push a trapped particle through the interface.
3. Measure the trap forces using the escape velocity method.
4. Trap some living cells (baker's yeast or probiotic bacteria) or trap particles inside living cells (organelles in a sheet of onion cells or colloids in a chicken's egg).
5. Model the trapping forces using the MatLab toolbox

   https://uk.mathworks.com/matlabcentral/fileexchange/73541-ott-optical-tweezers-toolbox

6 SAFETY

The laser used in this experiment is NOT eye safe! If the direct collimated beam were to enter your eye, damage to your retina WOULD occur. However, the tweezer system is quite safe so long as you DO NOT try to access the collimated beam. The system is contained in an interlocked box so that the laser can only be operated with this box closed. DO NOT attempt to override the interlock.

Immersion oil for the objective lens is poisonous if ingested. Take care to wash your hands after using it.

References

[1] A Ashkin. Acceleration and trapping of particles by radiation pressure. *Physical Review Letters*, 24(4):156–159, January 1970.

[2] A Ashkin and J M Dziedzic. Optical levitation by radiation pressure. *Applied Physics Letters*, 19(8):283–285, December 1971.

[3] A Ashkin, J M Dziedzic, J E Bjorkholm, and Steven Chu. Observation of a single-beam gradient force optical trap for dielectric particles. *Optics Letters*, 11(5):288, 1986.

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